

Homework 3 Table and Questions

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Table of Mass Break Down of the Local Group

Galaxy Name	Halo Mass 10^{12} SolarMass	Disk Mass 10^{12} SolarMass	Bulge Mass 10^{12} SolarMass	Total 10^{12} SolarMass	f_{bar}
Milky Way	1.975	0.075	0.01	2.06	0.04126214
M31	1.921	0.12	0.019	2.06	0.06747573
M33	0.187	0.009	0.	0.196	0.04591837
Local Group				4.316	0.05398517145505089

Table 1: Last Row contains the Total Mass and Baryon fraction of the whole Local Group.

Answers to Part 4 of Homework 3

1) The total mass of the Milky Way and M31 are the same in this simulation. For both galaxies the Halo Mass dominates the total mass.

2) The stellar mass of M31 is greater than the stellar mass of the Milky Way. Would expect M31 to be more luminous.

3) The ratio of dark matter mass of the Milky Way to the M31 is $\frac{1.975}{1.921}$. This is surprisingly close to one another given the stellar mass is greater for M31. However, given the notion that dark matter makes up the majority of matter in the universe, it makes sense the ratio would be fairly close, just because dark matter is likely to dominate most mass breakdowns.

4)The Baryon fraction of the universe is much larger than in any of the galaxies. The only thing I can think of that would impact the difference between these fractions is the fact that the universal Baryon fraction includes the Intergalactic Medium.